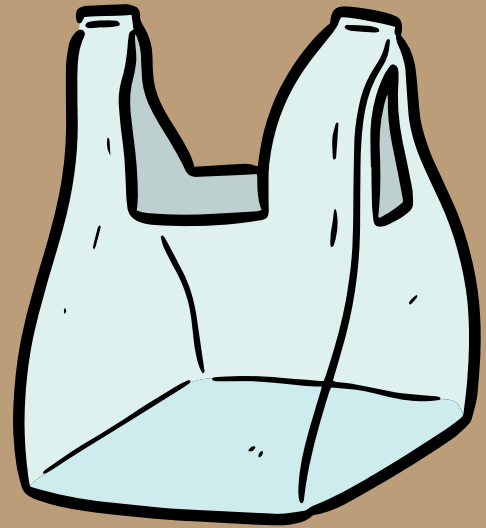




Life Cycle Impact Assessment²-



Carry Bags



GROUP 10

CHHABILA(MSCC1250017)

ANJANA (MSCC1250024)

ADITHYA (MSCC1250012)

Project case study #11: LCA of shopping bags

STUDY CONTEXT:

An LCA analysis of an average shoppers need for shopping bags that is made of

- Plastic (PE)- 15 Bags
- Cloth- 1 Bag
- Paper- 15 Bags

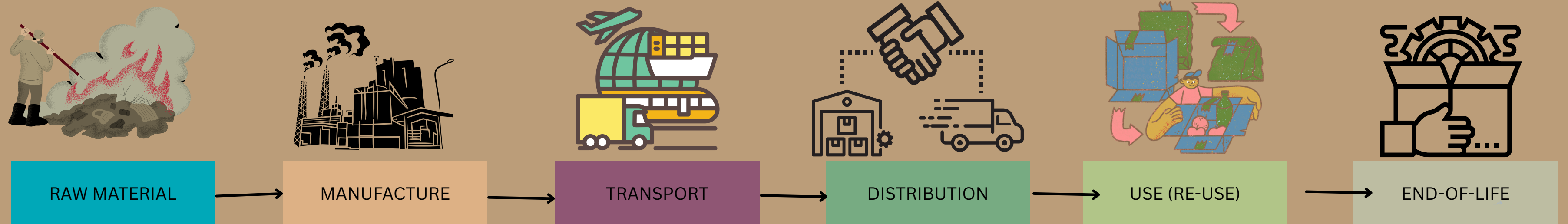
A cloth bag is used 50 times, whereas the plastic and paper are used only once.

EXCLUDED

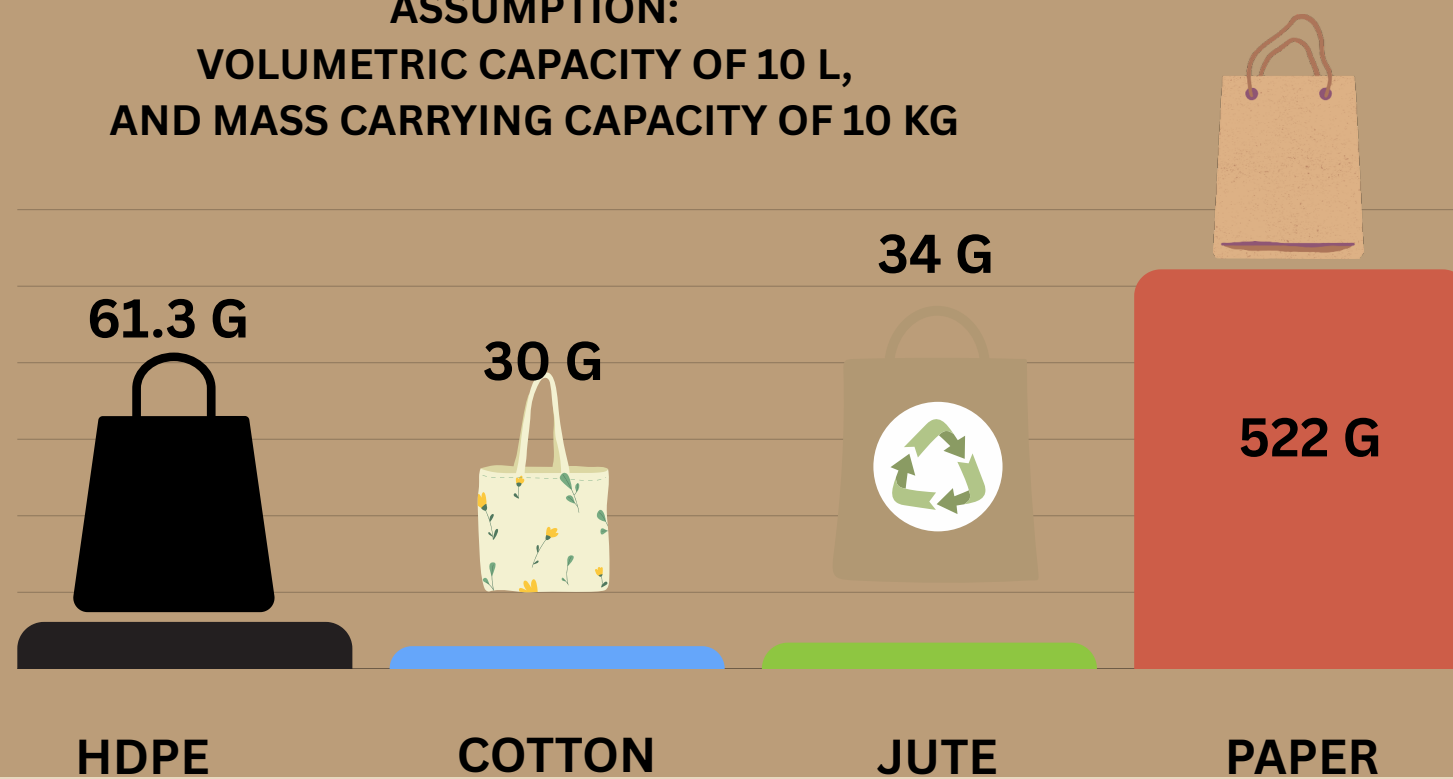
- Capital Cost
- Secondary Packing
- Litter

GOAL AND SCOPE OF THE STUDY

CRADLE -TO-GRAVE LCA: CARRY BAG (INDIA)



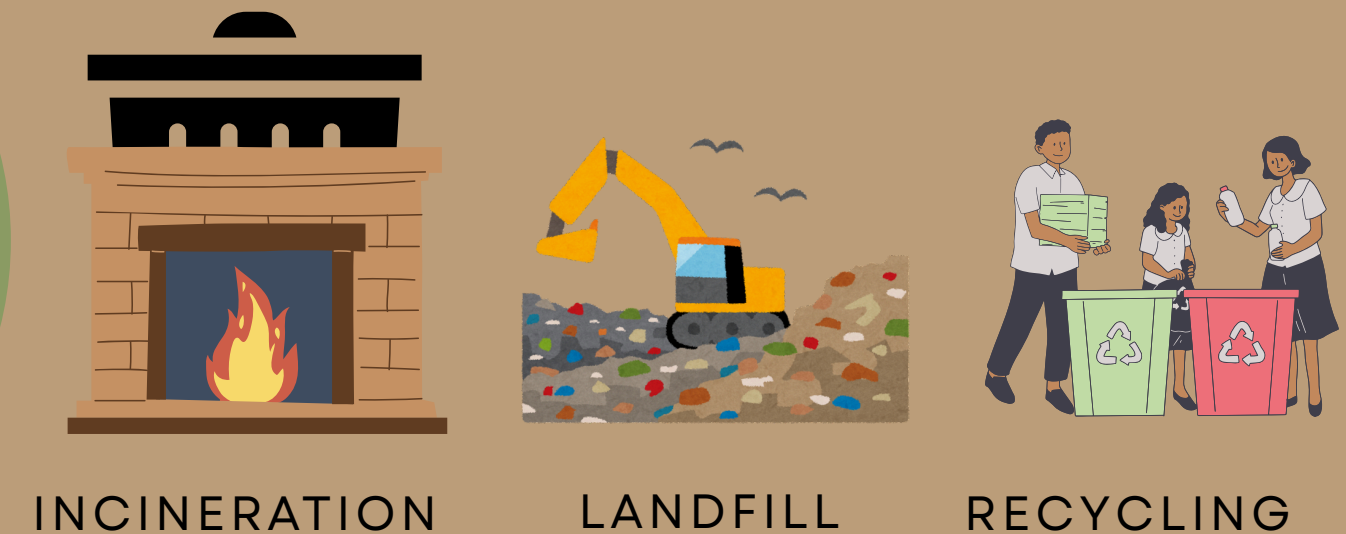
ASSUMPTION:
VOLUMETRIC CAPACITY OF 10 L,
AND MASS CARRYING CAPACITY OF 10 KG



FUNCTIONAL UNIT:

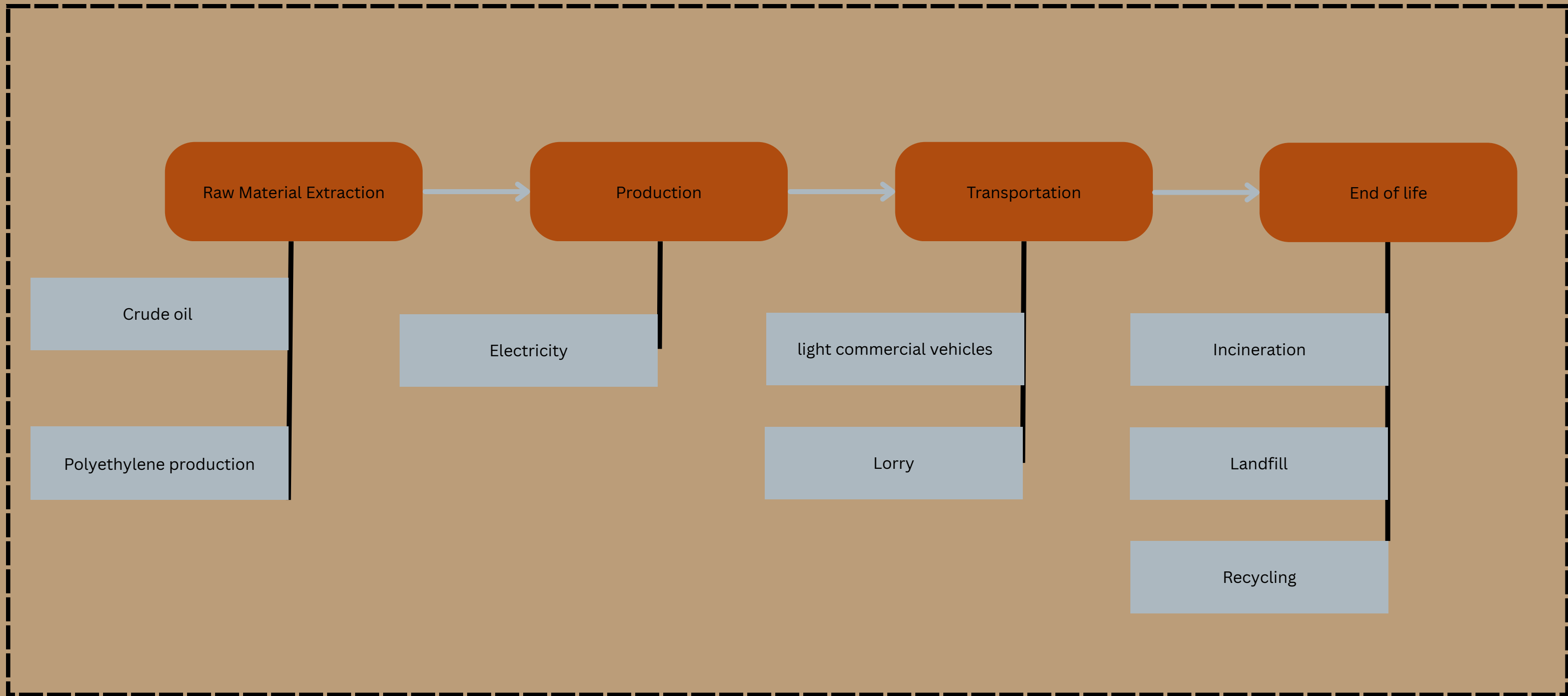
15
SUPERMARKET TRIPS
(YEAR./ HOUSEHOLDS)

3-SCENARIOS



TO COMPARE ENVIRONMENTAL IMPACTS

System Boundary Plastic Bag

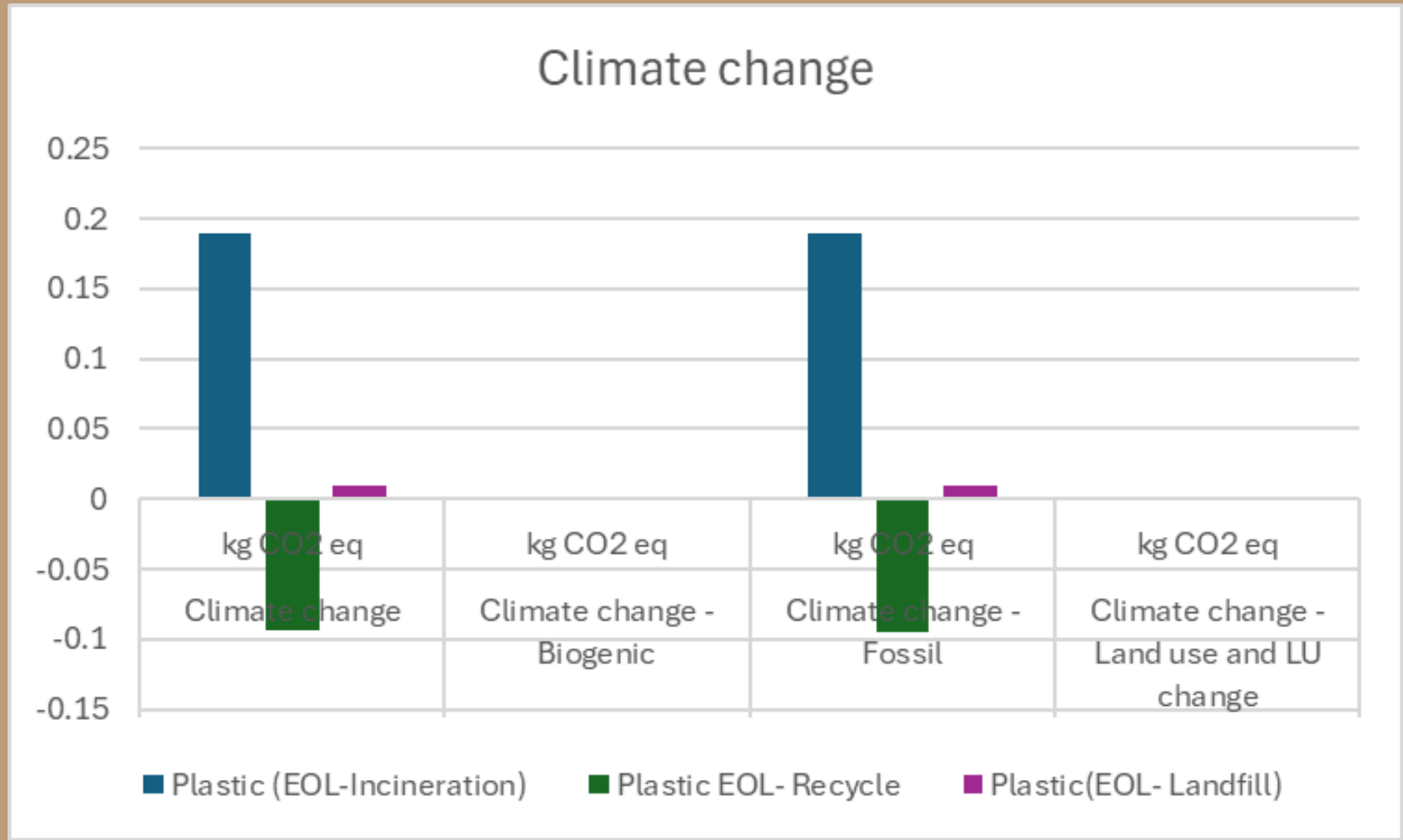


LCA Inventory

Plastic Bag

Raw materials		
Polyethylene, high density, granulate {RoW} polyethylene production, high density, granulate Cut-off, U	61.3	g
Manufacturing		
Extrusion, plastic film {RoW} extrusion, plastic film Cut-off, U	61.3	g
Transportation		
Transport, freight, light commercial vehicle, fleet average {RoW} transport, freight, light commercial vehicle, fleet average Cut-off, U	(61.3*50)/1000000	tkm
Transport, freight, lorry, >32 metric ton, diesel, EURO 6 {RoW} transport, freight, lorry, >32 metric ton, diesel, EURO 6 Cut-off, U	(61.3*50)/1000000	tkm
EOL- Incineration, Recycle, Land fill		
Waste polyethylene {, GLO} treatment of waste polyethylene, municipal incineration Cut-off, U.	61.3	g
PE (waste treatment) {GLO} recycling of PE Cut-off, U	61.3	g
Waste polyethylene {GLO} treatment of waste polyethylene, open dump, dry infiltration class (100mm) Cut-off, U	61.3	g

Plastic Bag

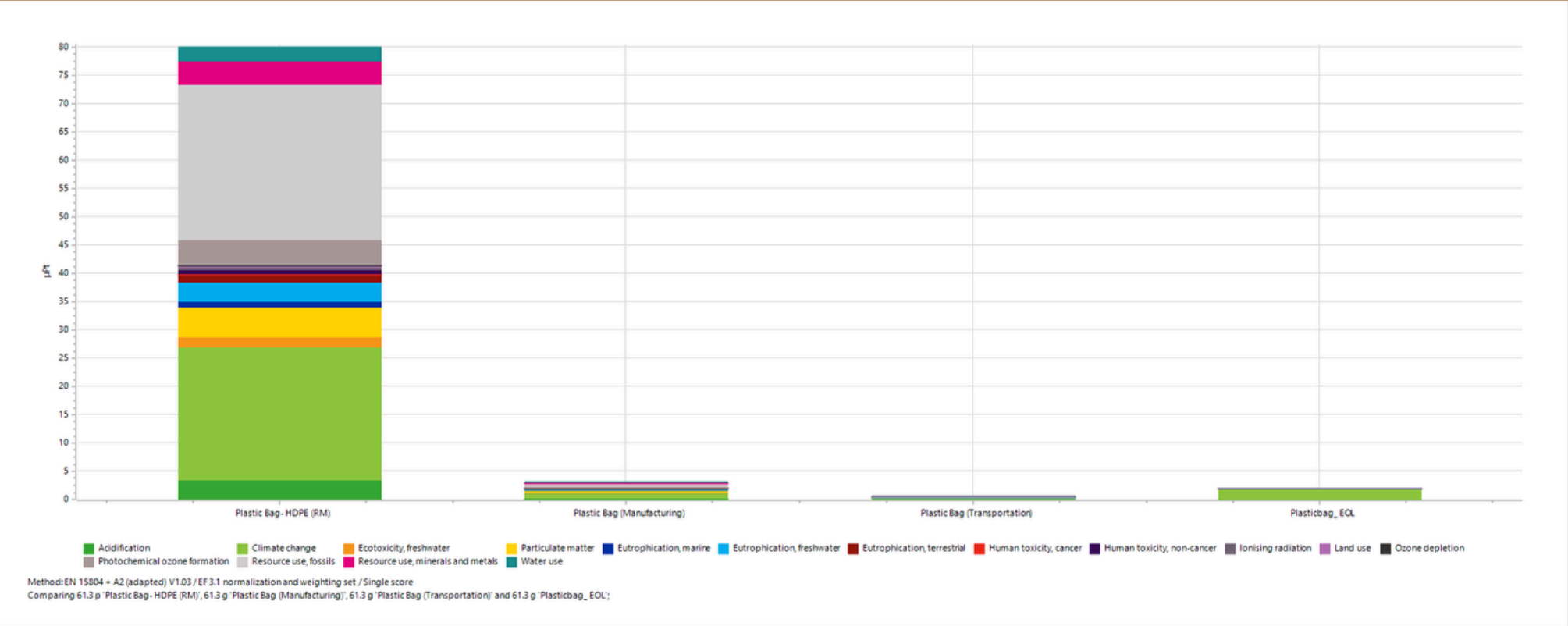


Impact

Plastic bag raw material has higher impact followed by manufacturing and end of life. transportation has least impact. for all the impact category.

EOL

EOL- Incineration of Plastic bag has the highest impact



LCA Inventory

PAPER BAG

Raw Materials



Edit material process 'Paper Bag (Adi_Raw Materials)'		
Documentation	Input/output	Parameters
Add line		
Inputs from technosphere: materials/fuels		Amount Unit
Kraft paper {RoW} kraft paper production Cut-off, U		38.6 g
Polyurethane adhesive {GLO} polyurethane adhesive production Cut-off, U		1 g



Manufacturing

Edit material process 'Paper Bag (Adi_Manufacturing)'		
Documentation	Input/output	Parameters
Add line		
Inputs from technosphere: electricity/heat		Amount Unit
Electricity, medium voltage {IN-Western grid} electricity voltage transformation from high to medium voltage Cut-off, U		0.056 kWh
Add line		



Transport

Edit material process 'Paper Bag_3(Adi_Transport)'		
Documentation	Input/output	Parameters
Add line		
Inputs from technosphere: electricity/heat		Amount Unit
Transport, freight, train, fleet average {RoW} transport, freight, train, electric Cut-off, U		$(1500 \times 34.8) / 1000000 = 0.0522$ tkm
Transport, freight, lorry, 16-32 metric ton, diesel, EURO 3 {RoW} transport, freight, lorry, 16-32 metric ton, diesel, EURO 3 ...		$(200 \times 34.8) / 1000000 = 0.00696$ tkm

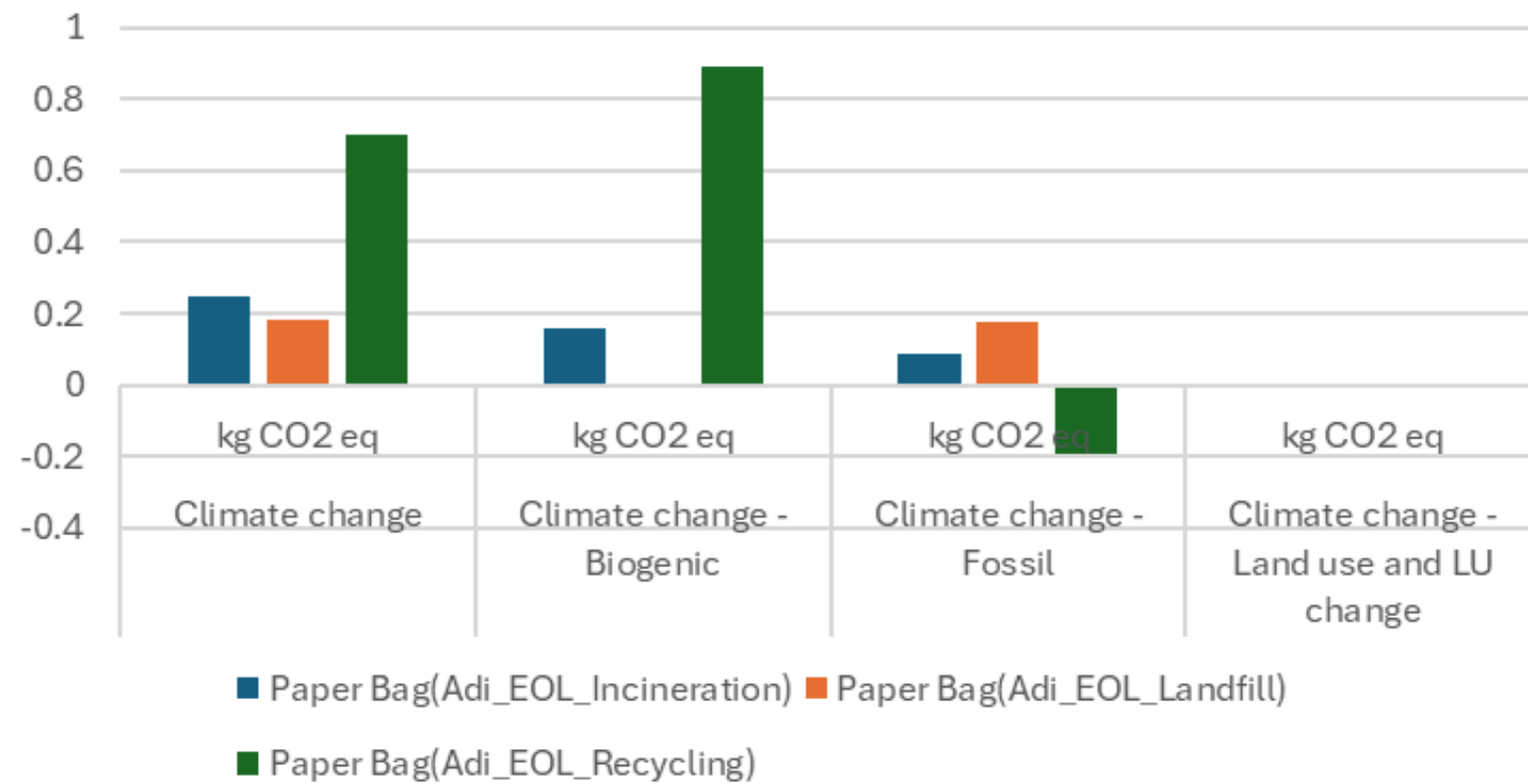


End-of-life

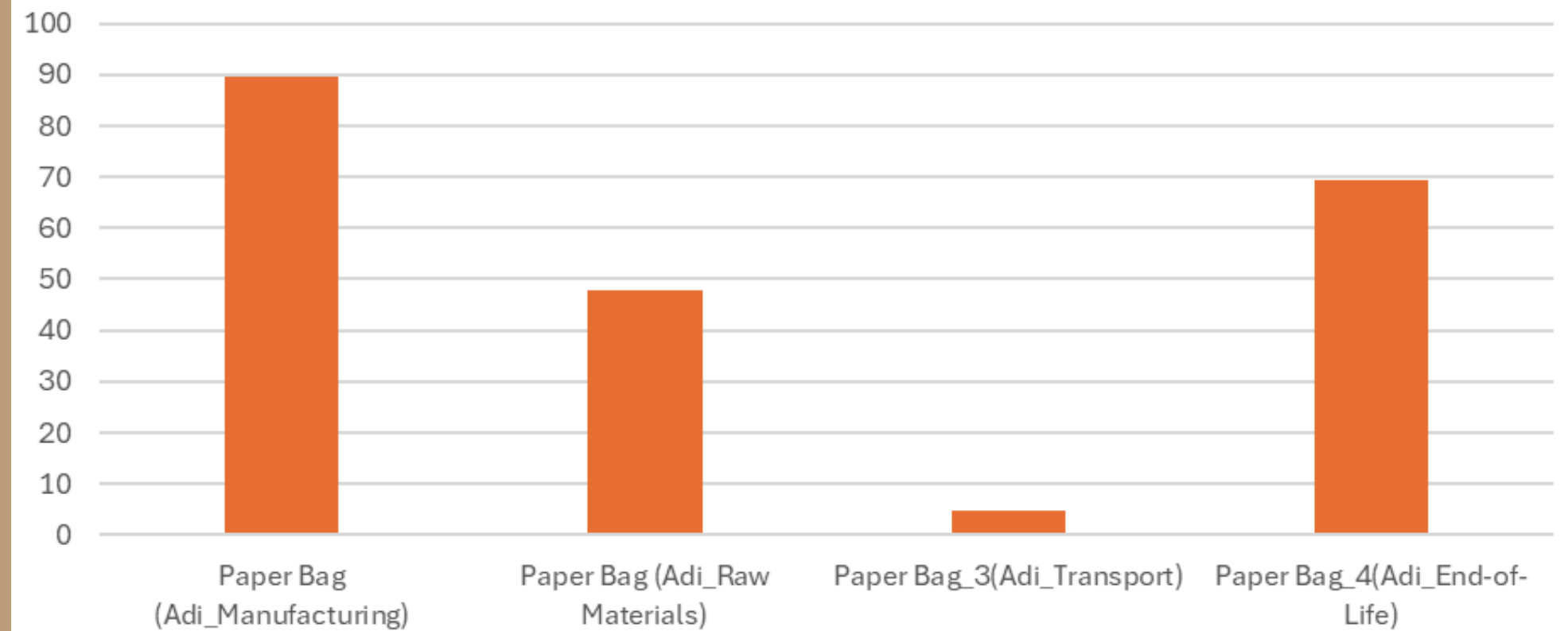
Edit material process 'Paper Bag_4(Adi_End-of-Life)'		
Documentation	Input/output	Parameters
Add line		
Municipal solid waste {RoW} treatment of municipal solid waste, sanitary landfill Cut-off, U		156.6 g
Paper (waste treatment) {GLO} recycling of paper Cut-off, U		104.4 g
Municipal solid waste {RoW} treatment of municipal solid waste, open burning Cut-off, U		261 g

Paper Bag

Climate change



Lfe Cycle Assessment of Paper Bag

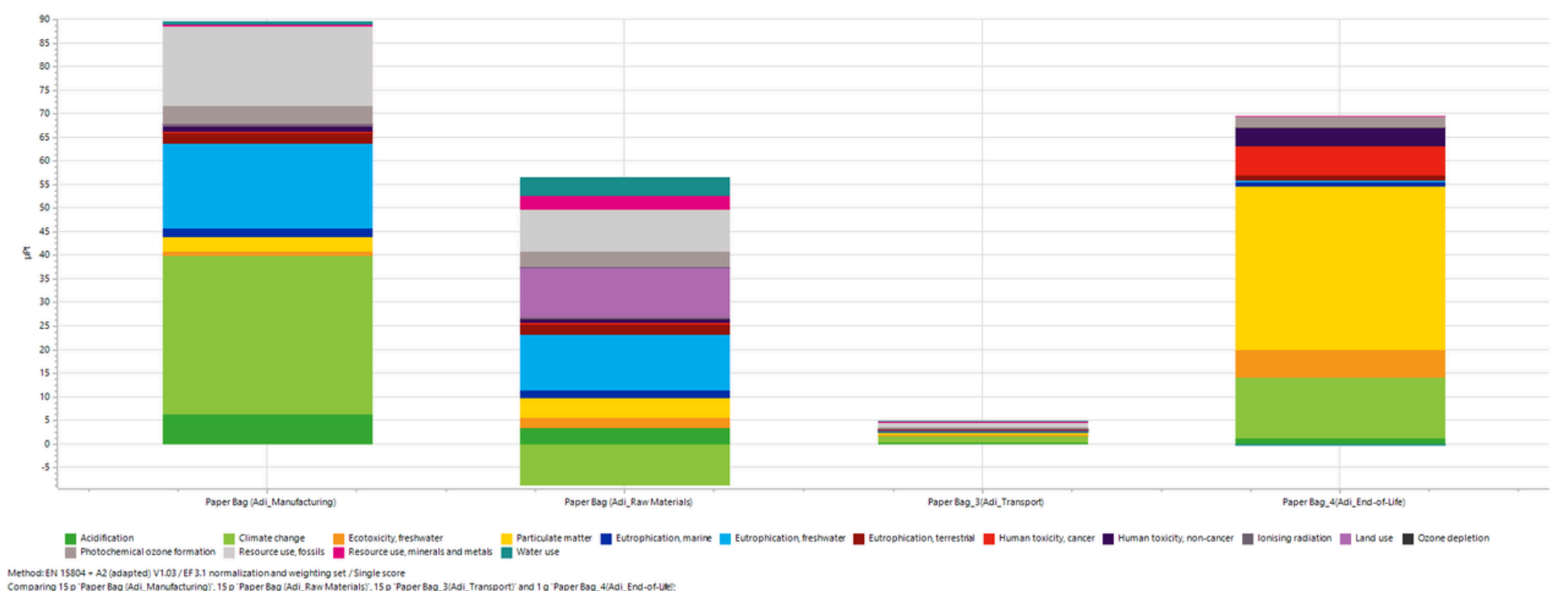


Impact

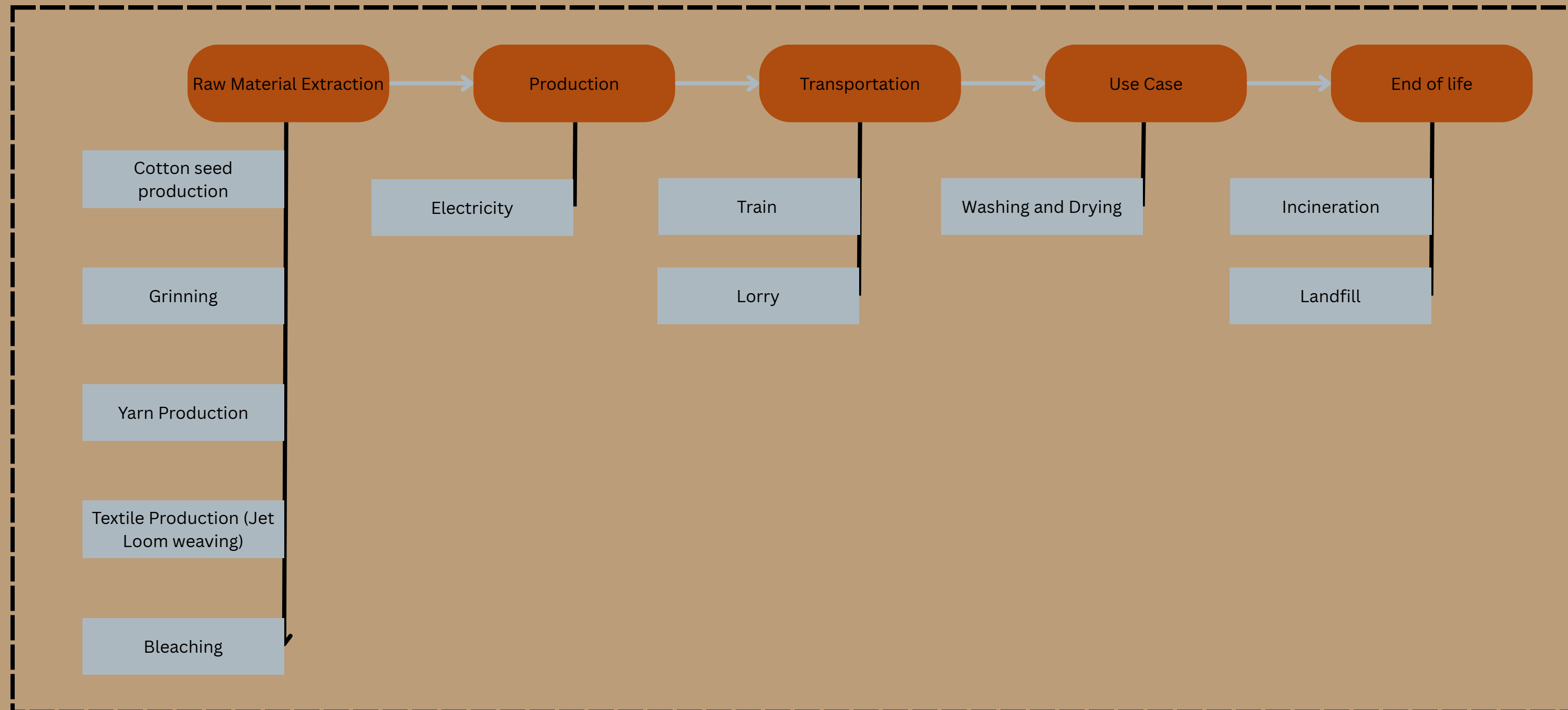
Paper bag manufacturing has higher impact followed by Raw material and Transport for all the impact category except for the freshwater eutrophication

EOL

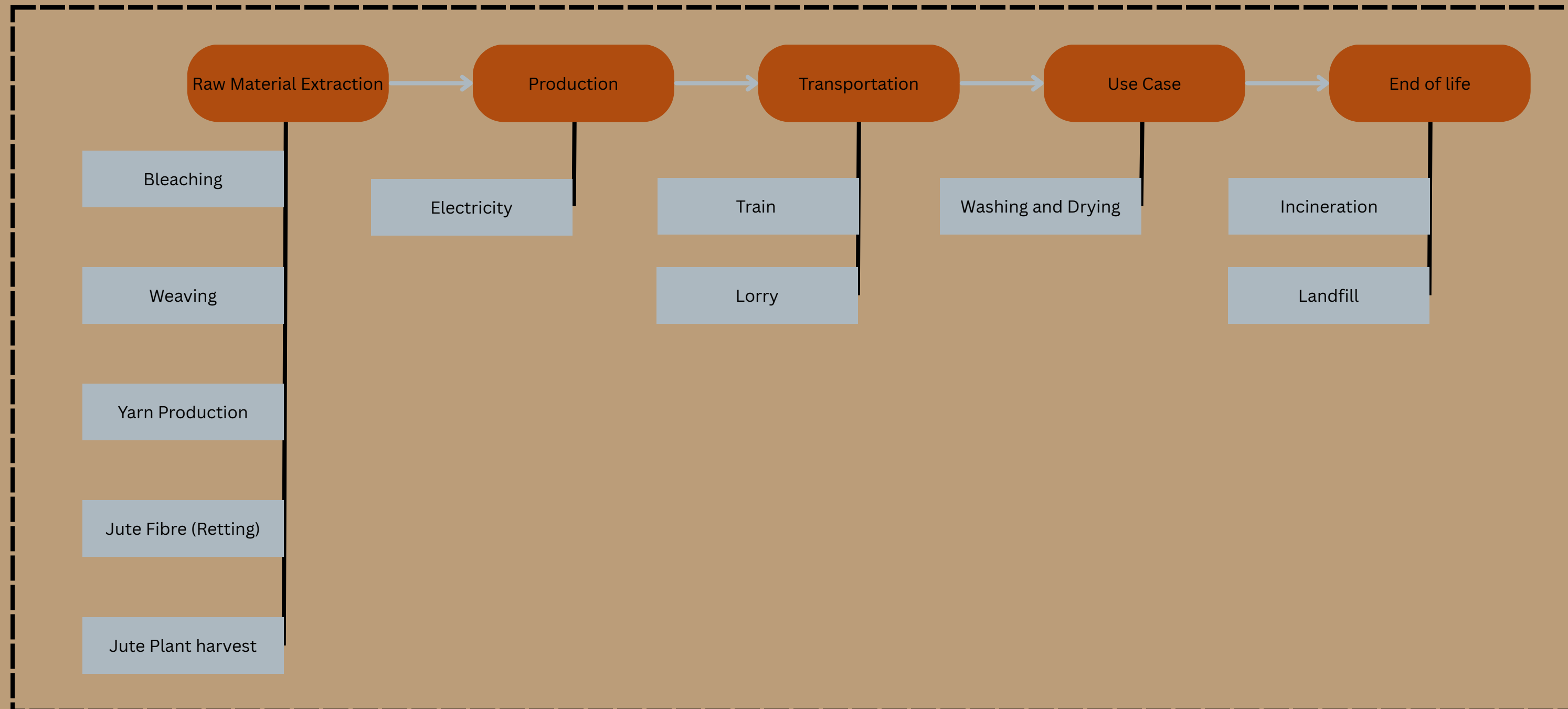
EOL- Recycling of Paper bag has the highest total impact



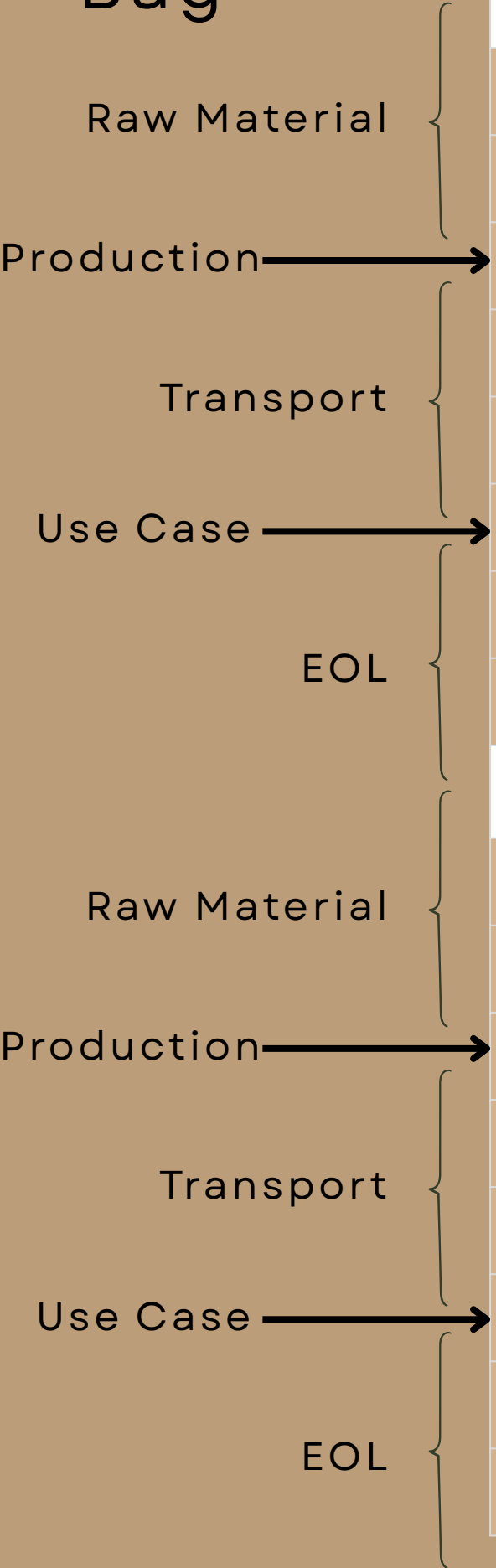
System Boundary



System Boundary

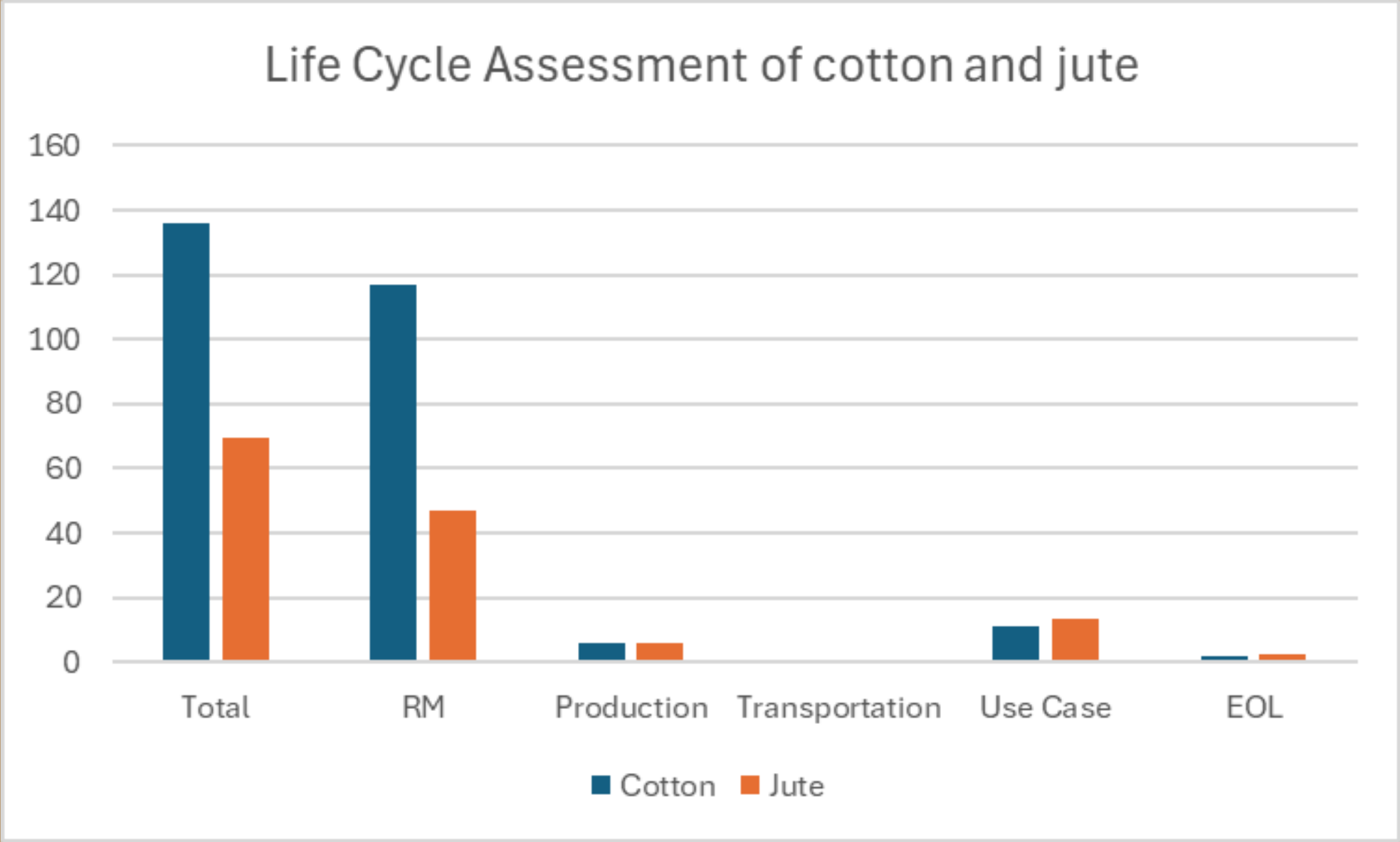
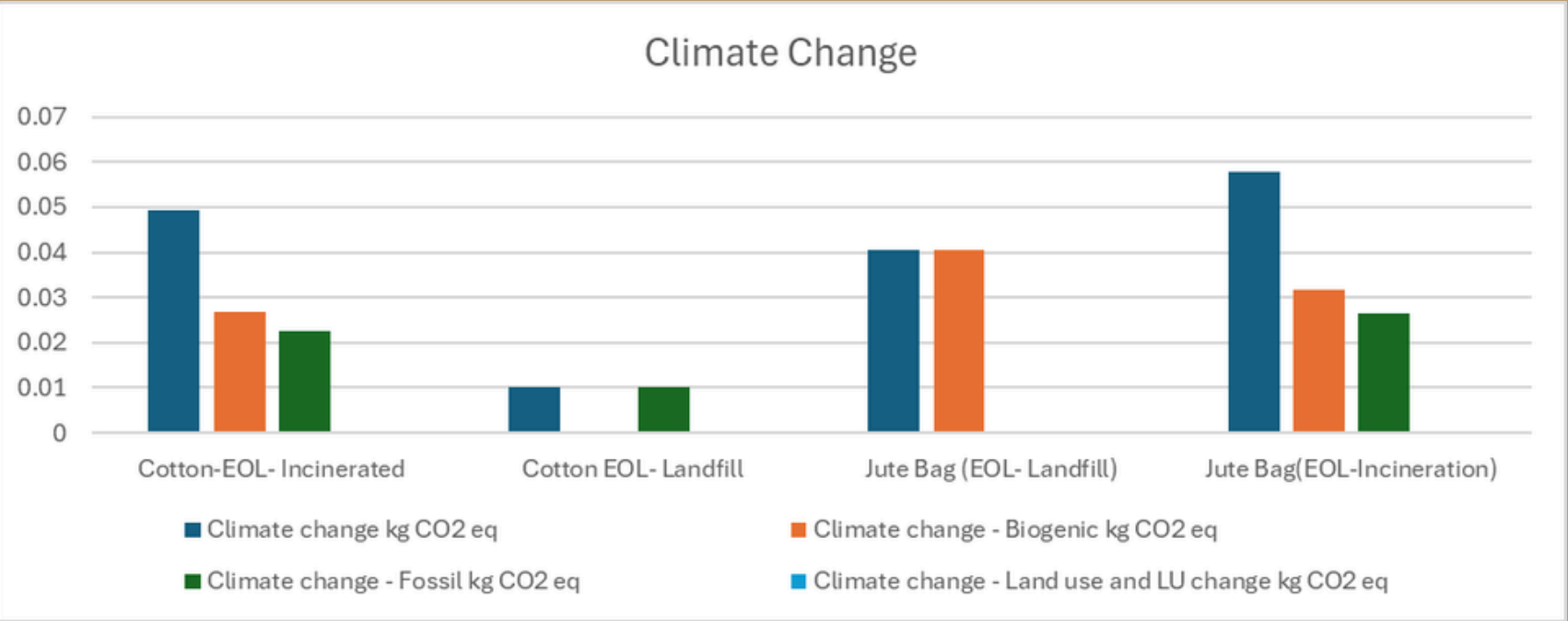


Inventory Cloth Bag



Cotton Bag		
Bleaching and dyeing, yarn {IN} bleaching and dyeing, yarn Cut-off, U	32.12	g
Textile, woven cotton {IN} textile production, cotton, air jet loom weaving Cut-off, U	33.81	g
Electricity, medium voltage {IN-Western grid} electricity voltage transformation from high to medium voltage Cut-off, U	0.2	MJ
Transport, freight, train, fleet average {IN} transport, freight, train, electric Cut-off, U	(1800*29.6)/1000000	tkm
Transport, freight, lorry, 3.5-7.5 metric ton, diesel, EURO 6 {RoW} transport, freight, lorry, 3.5-7.5 metric ton, diesel, EURO 6 Cut-off, U	(200*29.6)/1000000	tkm
Washing, drying and finishing laundry {GLO} washing, drying and finishing laundry Cut-off, U	29.6*15	g
Residues, MSWI, waste textile, soiled {GLO} treatment of residues, MSWI, waste textile, soiled, residual material landfill Cut-off, U	29.6	g
Waste textile, soiled {GLO} treatment of waste textile, soiled, municipal incineration Cut-off, U	29.6	g
Jute Bag		
Textile, jute {IN} textile production, jute, weaving Cut-off, U	39.7	g
Bleaching, textile {IN} bleaching, textile Cut-off, U	37.7	g
Electricity, medium voltage {IN-Eastern grid} electricity voltage transformation from high to medium voltage Cut-off, U	0.2	MJ
Transport, freight, train, fleet average {IN} transport, freight, train, electric Cut-off, U	(1800*34.8)/1000000	tkm
Transport, freight, lorry, 3.5-7.5 metric ton, diesel, EURO 6 {RoW} transport, freight, lorry, 3.5-7.5 metric ton, diesel, EURO 6 Cut-off, U	(200*34.8)/1000000	tkm
Washing, drying and finishing laundry {GLO} washing, drying and finishing laundry Cut-off, U	34.8*15	g
Waste textile, soiled {GLO} treatment of waste textile, soiled, municipal incineration Cut-off, S	34.8	g
Waste yarn and waste textile {IN} treatment of waste yarn and waste textile, unsanitary landfill Cut-off, U	34.8	g

CloTh Bag

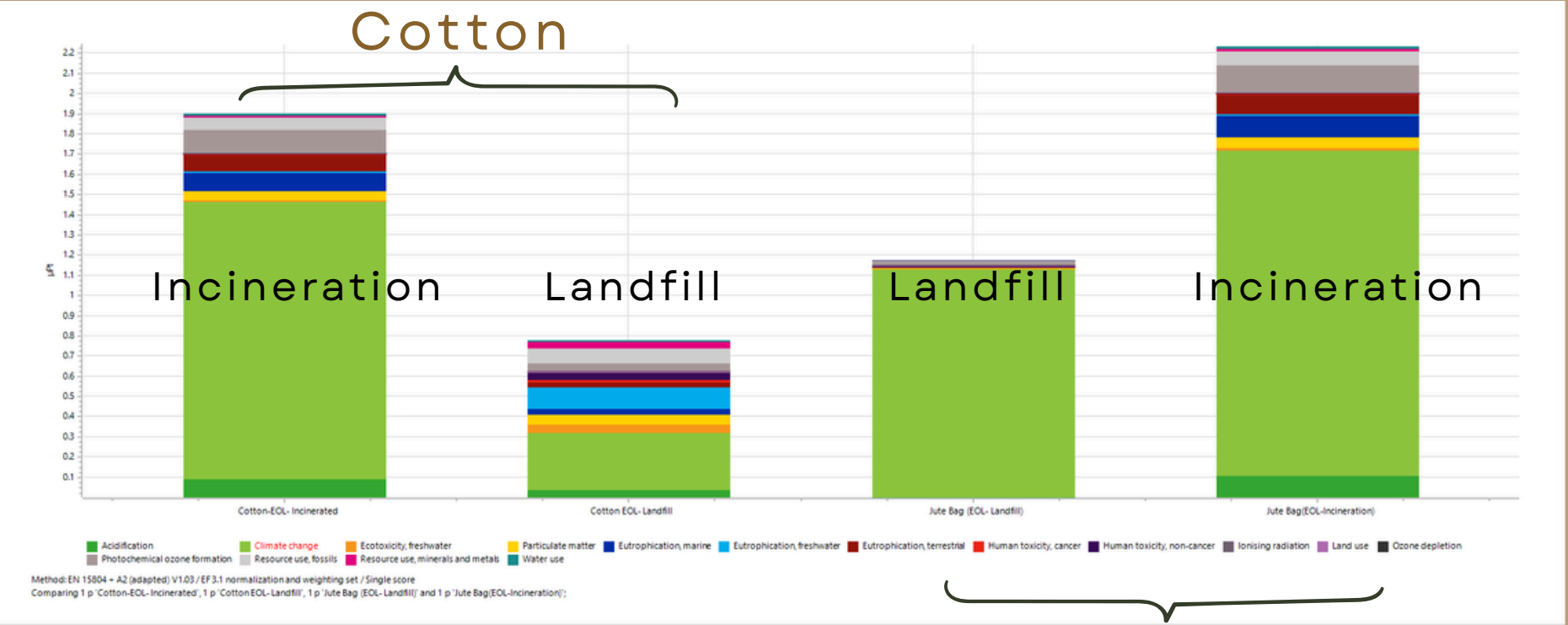


Impact

Cotton has higher impact for all the impact category except for the freshwater eutrophication

EOL

EOL- Incineration of cotton and jute has the highest total impact

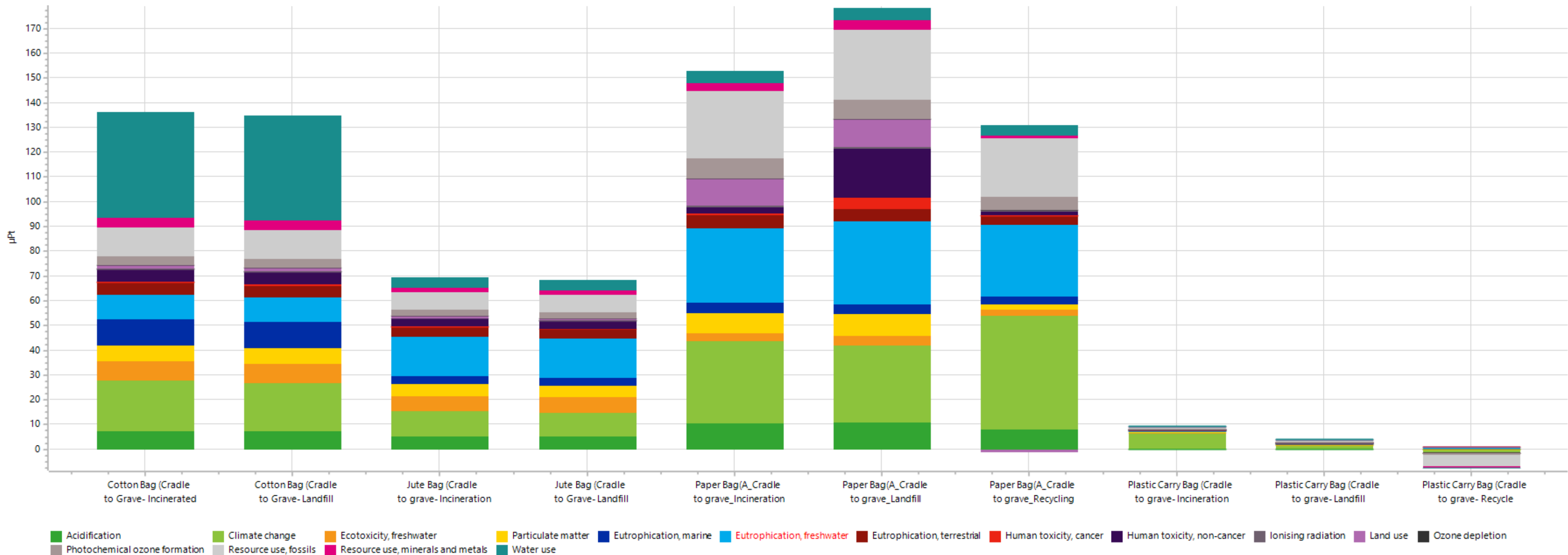


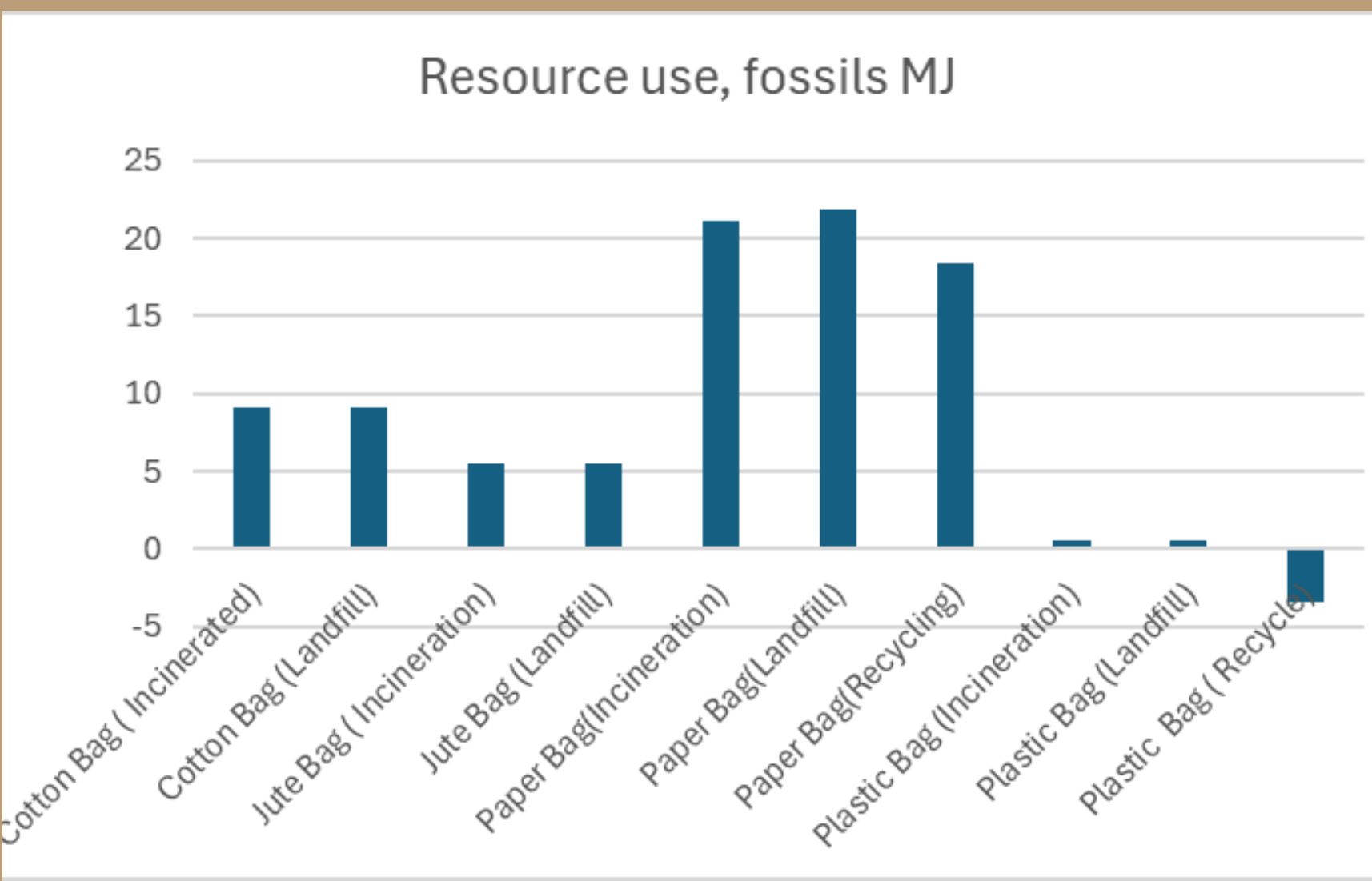
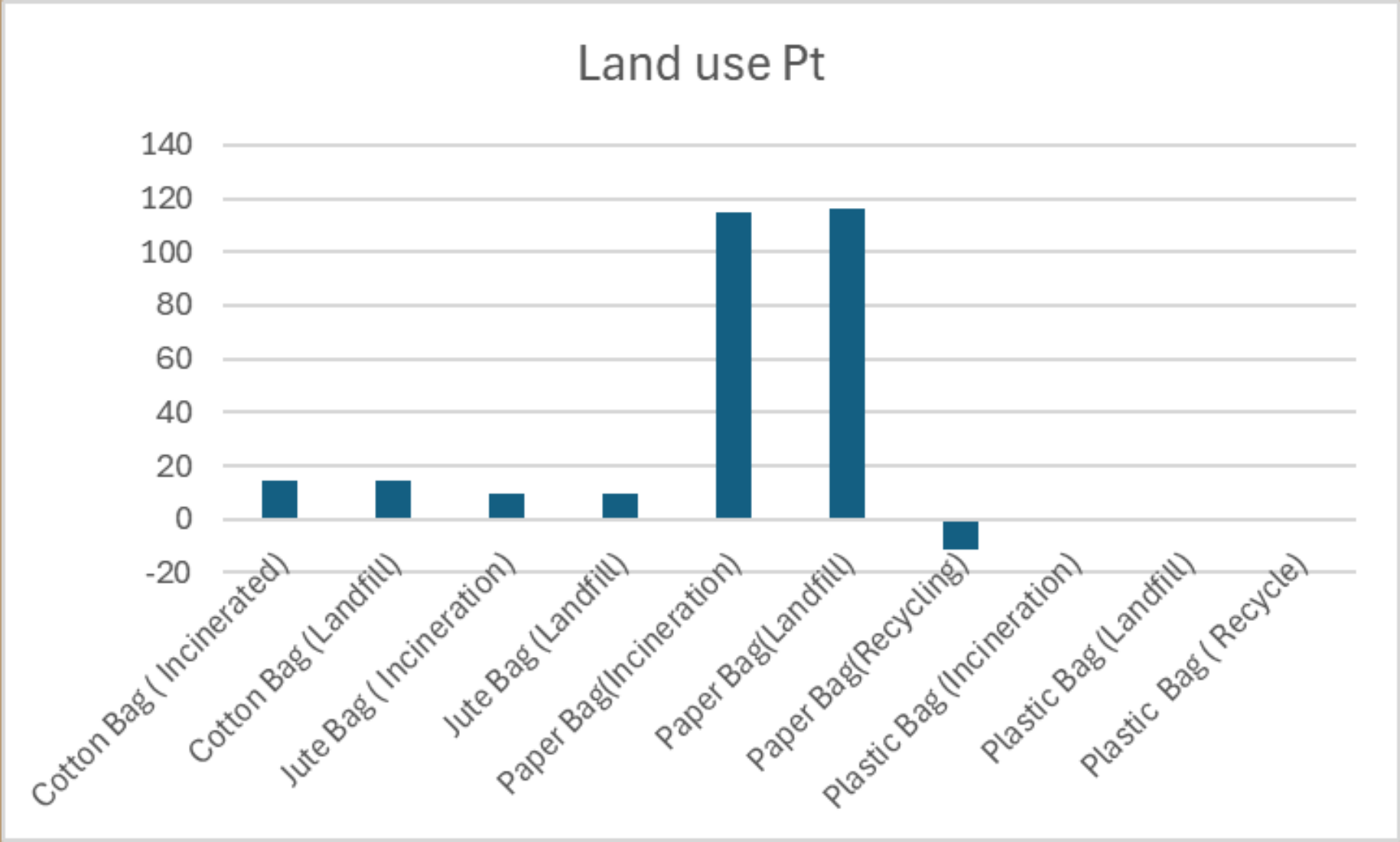
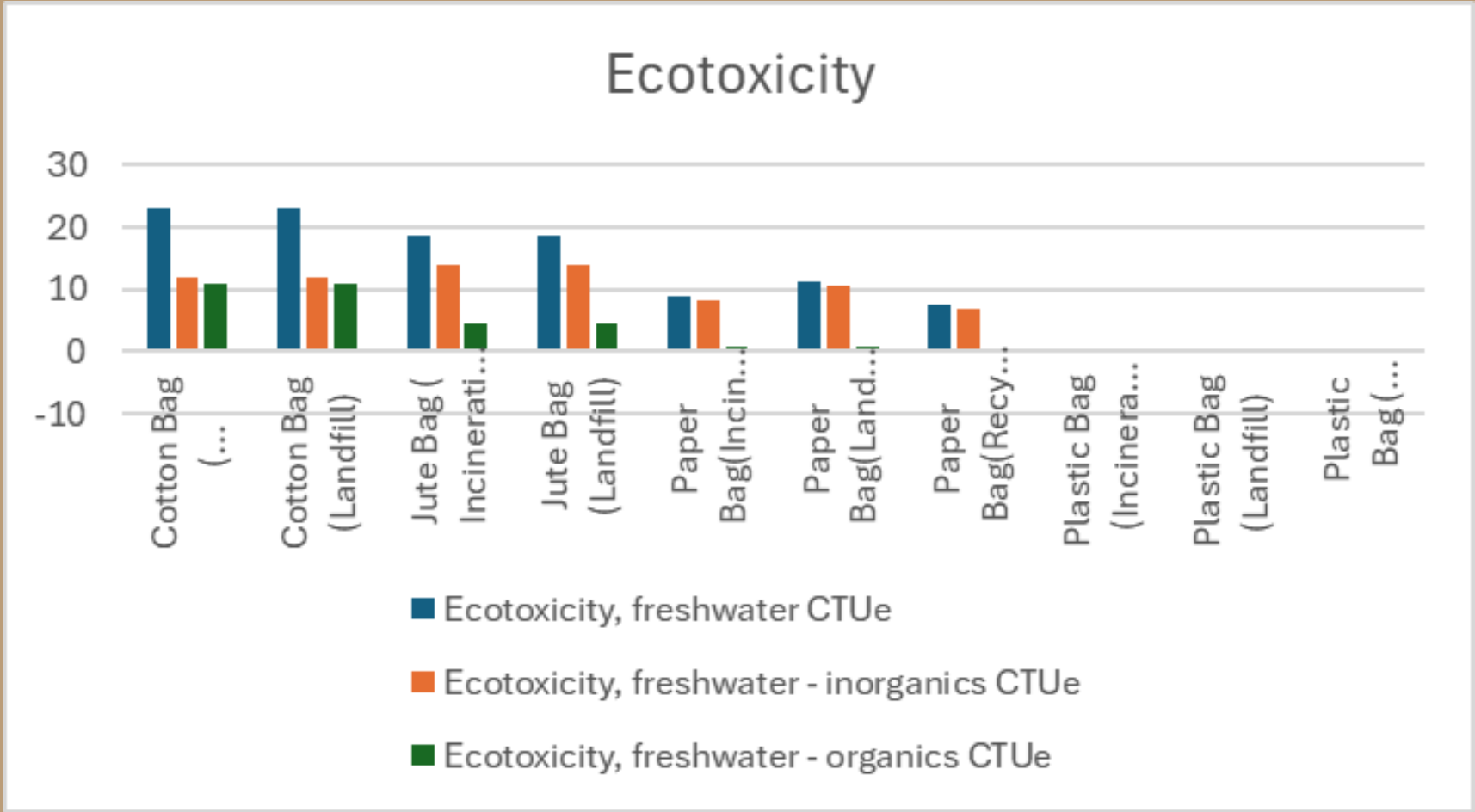
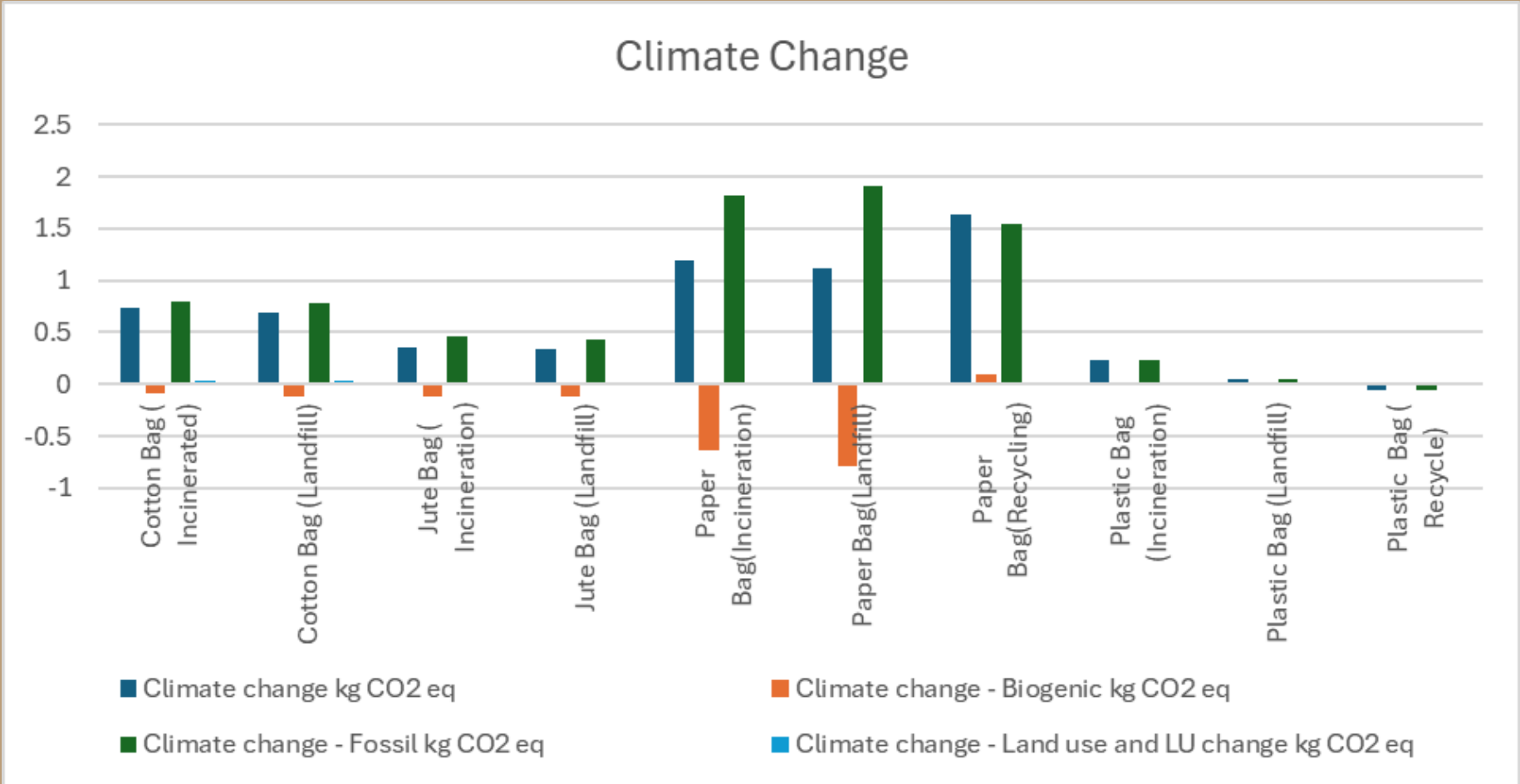
Results

Single Score

Refer Dashboard

The comparison of the different carry bags shows that plastic has the least impact and further





Discussion

Cloth Bag

- Cotton and jute have a higher impact on Raw materials
- Cotton RM:**
- Of the total climate change of 0.732 Kg CO₂ eq, Raw material alone constitutes around **63%**.
- Water consumption constitutes around 99.2% of 5.73 m³
- Jute:**
- Raw material constitutes 15.3 % and use case constitutes 43.2% of the total climate change, because of the energy required for washing the bag.
 - Water use is higher for Raw material, which is 90.1%

Jute is better than cotton

Paper Bag

- Manufacturing: 1.20 kg CO₂ eq → moderate
 - Raw Materials: 0.31 kg CO₂ → eq moderate to low
 - Transport: 0.05 kg CO₂ eq → negligible
- End-of-Life stage; disposal method decides overall footprint.**
- Incineration: 0.245 kg CO₂ eq
 - Landfill: 0.180 kg CO₂ eq → lowest impact
 - Recycling: 0.699 kg CO₂ eq → highest impact
-  From a climate change perspective:
- Landfill performs best.
Incineration is moderately worse, and
Recycling is nearly 3x higher in impact relative to landfill.**

Plastic Bag

- **HDPE-** High density polyethylene shopping bag.
- Comparing to cloth and paper bag plastic bag have less impact on environment.
- Under the life cycle assessment of plastic bag raw material part is contributing major impact on environment.
- 80% of the total impact is coming from raw materials.
- All other phases- manufacturing, transportation, end of life have a less impact comparing other. each of these are contributing less than 5%.

Recycling has the lowest impact

Across all the categories, plastic is performing better except for the paper bag under the recycling scenario for the land use category

THANK YOU

